

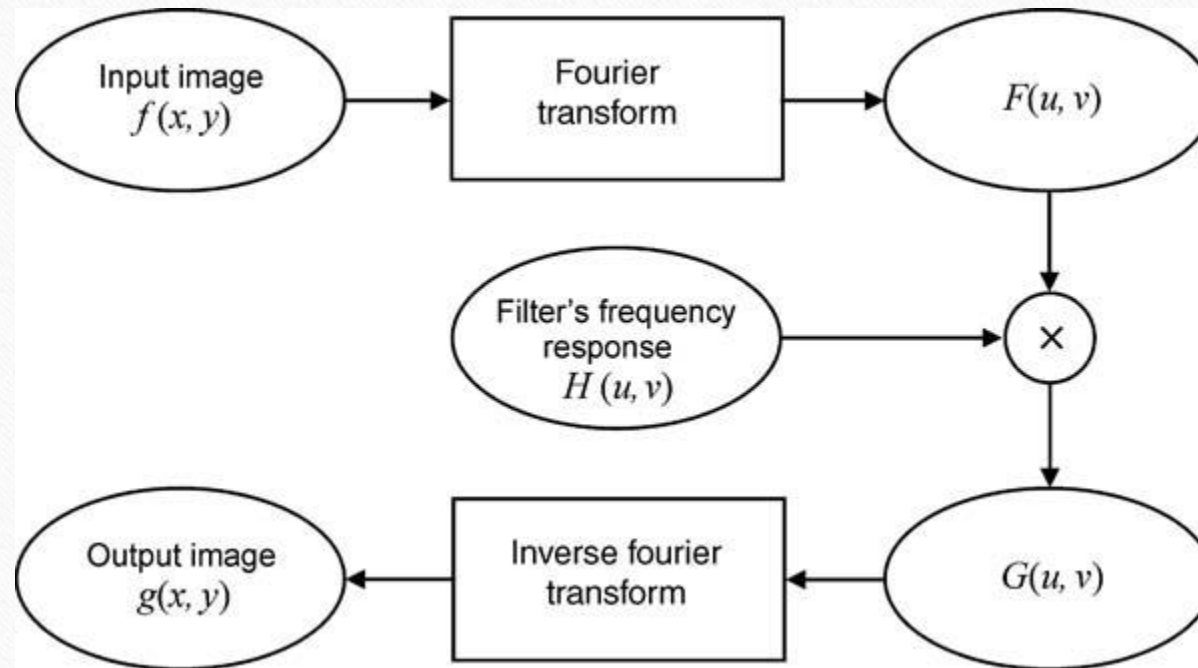
CHAPTER 11

FREQUENCY-DOMAIN FILTERING

WHAT WILL WE LEARN?

- Which mathematical tools are used to represent an image's contents in the 2D frequency domain?
- What is the Fourier transform, what are its main properties, and how is it used in the context of frequency-domain filtering?
- How are image processing filters designed and implemented in the frequency domain?
- What are the differences between low-pass and high-pass filters (HPFs)?
- What are the differences among ideal, Butterworth, and Gaussian filters?

Frequency-domain operations



TRANSFORM

- A *transform* is a mathematical tool that allows the conversion of a set of values to another set of values, creating, therefore, a new way of representing the same information.
 - Original domain *spatial domain*
 - Results domain: *transform domain*

Forward and Inverse transform

- Forward transform

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot r(x, y, u, v)$$

- Inverse transform

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) \cdot s(x, y, u, v)$$

The 2D Discrete Fourier Transform

The 2D Fourier transform of an image is a special case of equations (11.4) and (11.5) where

$$r(x, y, u, v) = \exp[-j2\pi(ux/M + vy/N)] \quad (11.6)$$

and

$$s(x, y, u, v) = \frac{1}{MN} \exp[j2\pi(ux/M + vy/N)] \quad (11.7)$$

where $j = \sqrt{-1}$.

In MATLAB

- 2D FT: *fft2*
- Inverse FT: *ifft2*
- Example:

```
I = imread('Figure11_04_a.png');  
Id = im2double(I);  
ft = fft2(Id);  
ft_shift = fftshift(ft);  
imshow(log(1 + abs(ft_shift)), [])
```

Properties of the Fourier Transform

- **Linearity:**

$$\mathfrak{F}[a \cdot f_1(x, y) + b \cdot f_2(x, y)] = a \cdot F_1(u, v) + b \cdot F_2(u, v)$$

- **Translation:**

$$a \cdot f_1(x, y) + b \cdot f_2(x, y) = \mathfrak{F}^{-1}[a \cdot F_1(u, v) + b \cdot F_2(u, v)]$$

$$\mathfrak{F}[f(x - x_0, y - y_0)] = F(u, v) \cdot \exp[-j2\pi(ux_0/M + vy_0/N)]$$

$$f(x - x_0, y - y_0) = \mathfrak{F}^{-1}F(u, v) \cdot \exp[j2\pi(ux_0/M + vy_0/N)]$$

- **Conjugate Symmetry:** $F(u, v) = F^*(-u, -v)$
- **Periodicity:** infinitely periodic in both the u and v directions
- **Separability:** can be computed by two passes of the 1D FT algorithm
- **Rotation:** If an image is rotated by a certain angle θ its 2D FT will be rotated by the same angle

Other Mathematical Transforms

LOW-PASS FILTERING

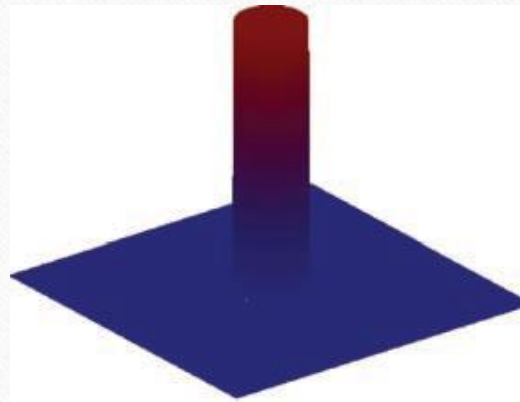
- Attenuate the high-frequency components and leaving the low-frequency components unchanged
 - Ideal LPF
 - Gaussian LPF
 - Butterworth LPF

Ideal LPF

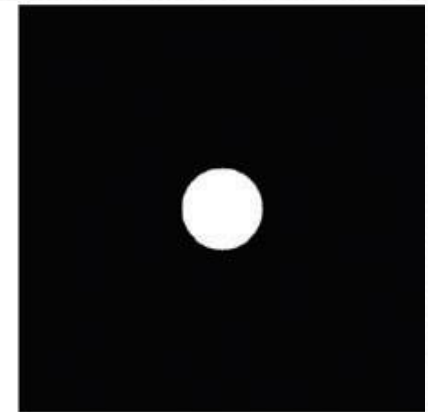
$$H_I(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

where $D(u, v) = \sqrt{u^2 + v^2}$
is distance between a point
of coordinates (u, v) and the
origin of the 2D frequency
plan

D_0 : *cutoff radius*



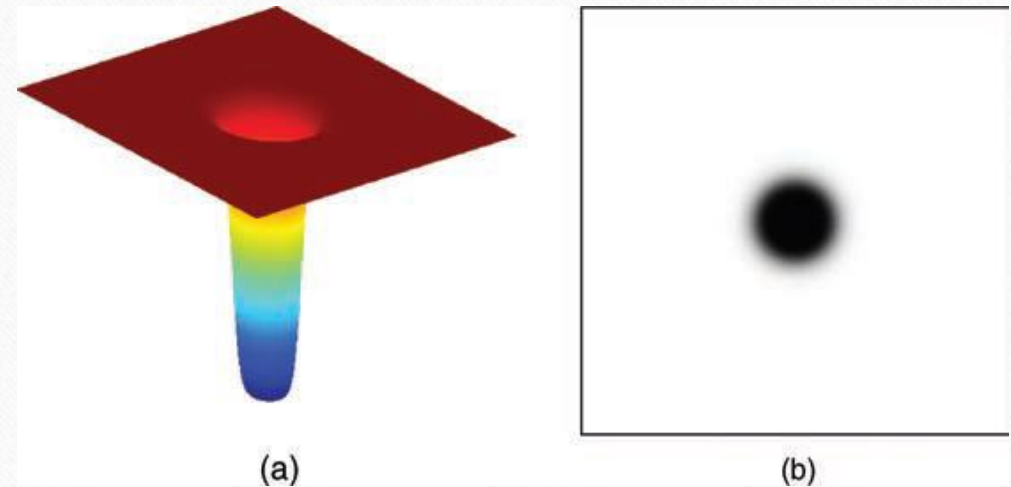
(a)



(b)

HIGH-PASS FILTERING

- Attenuate the low-frequency components
- Enhancing the high-frequency components



High-Frequency Emphasis

- Preserves the low-frequency contents of the input image (while enhancing its highfrequency components) by multiplying the high-pass filter function by a constant and adding an offset to the result

$$H_{\text{hfe}}(u, v) = a + bH(u, v)$$

Hands-on

- **TUTORIALS and PROBLEMS**